

A quick scenario

Email autocomplete feels easy forward, awkward backward

Typing an email (left → right)

User types: Please find the

System suggests: attached, updated, report

attached updated report

Reconstructing prior words (right ← left)

Cursor before: ... the | meeting

System guesses: before, during, for the

before during for the

- User action: draft an email; cursor moves forward or back.
- System: offers next-word vs prior-word suggestions.
- Asymmetry: forward feels fluent; backward feels uncertain.

Takeaway → Do models learn language better forward than backward?

Arrows of Time in LLMs

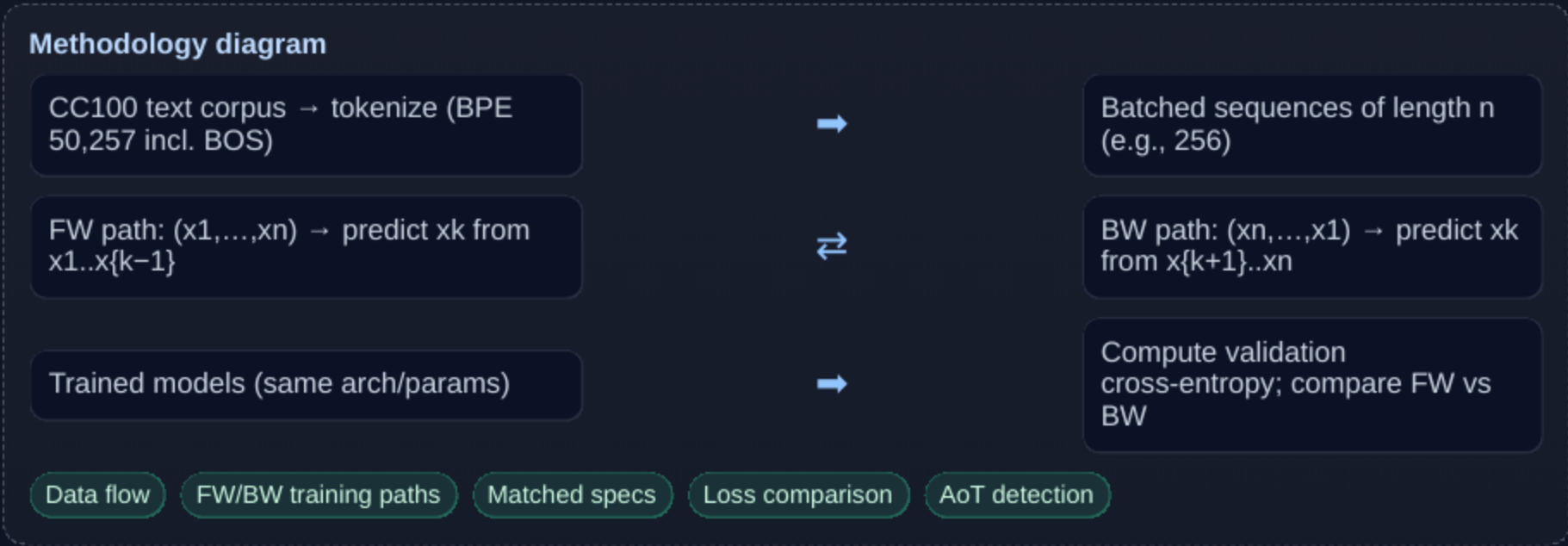
Why left-to-right autocomplete feels easier than right-to-left

- Do autoregressive LLMs reflect this asymmetry when trained forward vs backward?

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Setup and Method

- Same dataset, model, and hyperparameters; only token order differs (FW: next-token, BW: previous-token).
- Measure cross-entropy/perplexity to compare learned measures $P \rightarrow$ vs $P \leftarrow$.



Key Empirical Results

- FW AoT across languages and scales: BW loss persistently > FW.
- English/French (GPT2-Medium, n=256, ~27B tokens):
 - English: FW 2.880; BW +0.76%.
 - French: FW 2.788; BW +2.65%.
- Effect grows with context length (16 → 512) and with model size (~5M → 405M).
- Observed with GPT, GRU, LSTM; language-dependent magnitude (≈0.6%–3%+).
- Controls: tokenizer direction and init variance do not explain the gap.

Theory Intuition

- Compute limits break FW/BW symmetry predicted by information theory.
- Factoring dataset: multiplication (FW) easy; inversion (BW) hard → large loss gap.
- Linear languages: sparse forward maps have dense inverses → FW easier to learn than BW.
- Communication view: languages biased toward sparse FW updates; BW becomes denser.

Two mechanisms: (1) representation limits in BW; (2) learnability/optimization gaps.

Conclusions and Contact

- LLMs show a consistent FW Arrow of Time: FW perplexity < BW under matched training.
- Magnitude increases with context and capacity; varies by language.
- Complexity and sparsity offer a grounded explanation; synthetic tasks reproduce AoT.

Open directions

- Universality across languages/modalities; scaling laws; causality/V-information links.
- AoT in continuous data (video); relation to other arrows (thermodynamics).

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